

2 State of the Art

2.1 AI approaches in ASAG

Automatic short answer grading (ASAG) systems increasingly rely on a range of artificial intelligence (AI) techniques, from classical machine and deep learning to modern large language models (LLMs). The most notable technologies include Machine Learning (ML), Deep Learning (DL), Natural Language Processing (NLP), Recurrent Neural Networks (RNN), Bidirectional Encoder Representations from Transformers (BERT), Extra-Large Bidirectional Transformer Networks (XLNET), LLAMAs, and ChatGPT. This section summarises how these techniques have been integrated into ASAG, [18].

2.1.1 Deep Learning in ASAG

DL has significantly improved automatic grading by increasing accuracy and scalability, allowing educators to focus on higher-level teaching tasks. Studies show that RNNs, LSTMs, attention mechanisms, and transformer-based models dominate NLP for assessment tasks [19, 20, 21, 22, 23]. Response-based systems using deep neural networks outperform classical approaches [24, 22, 25, 26, 27], and have proven useful in both formative and summative contexts [28, 29, 30, 31]. Progress is largely due to transfer learning, fine-tuning massive pre-trained language models for ASAG [32, 33]. DL architectures also overcome traditional RNN limitations such as early-sequence information loss [34], making them central to modern ASAG pipelines [20, 21, 22].

2.1.2 NLP and Text Pre-processing

NLP enables machines to autonomously interpret free-text responses through ML/DL methods. Reported applications include machine translation [35], text sophistication evaluation [36], vocabulary development [37], social-network analysis [38], and essay scoring [34]. Publications on NLP-based auto-grading have steadily risen from 2016 to 2024 [39, 40, 41]. Text pre-processing: the initial stage in NLP, converts raw text into analyzable form via tokenisation, stop-word removal, normalisation, stemming/lemmatisation, and part-of-speech tagging. Tokenisation and stop-word removal focus the model on informative terms. Normalisation converts variants like “2moro” or “tomrw” to “tomorrow”. Lemmatisation provides linguistically correct base forms, while POS tagging supplies grammatical roles to improve semantic understanding. These steps enhance ML performance and model generalisation.

2.1.3 Recurrent Neural Networks (RNNs)

Unlike feed-forward networks, RNNs maintain hidden states capturing sequential dependencies, making them ideal for text analysis [42, 43]. RNN variants such as Gated Recurrent Units (GRU) and Long Short-Term Memory (LSTM) are the most widely employed [20, 44, 45]. They capture context and meaning in sentences, enabling ASAG systems to handle temporal order and dependencies effectively. Studies show rapid growth in RNN-based auto-grading research since 2016, across diverse subject areas.

2.1.4 Large Language Models (LLMs)

LLMs (e.g., BERT, GPT, LLAMA/LLAMA-2) have extended ASAG capabilities across disciplines [46, 47, 48, 49]. LLAMA-2 builds upon earlier LLMs with improved contextual understanding [50, 51]. These models are trained on large corpora and fine-tuned for grading tasks [48, 49, 52]. Despite their power, reliability concerns remain, particularly in high-stakes domains like medicine. Quah et al. evaluated GPT-3.5, GPT-4, Llama 2, Gemini, and Copilot, finding 68.9% accuracy on basic science and only 45.9% on pharmacology questions, advising use for education but not clinical decision-making [53]. Multilingual studies show mixed outcomes: Correa et al. demonstrated that English-based instructions cannot be directly applied to Portuguese [54], while cross-lingual comparisons across ChatGPT, Bing Chat, and Bard/Gemini revealed no clear leader [55]. In sentiment analysis across Czech, French, and English, LLAMA-2 and ChatGPT slightly outperformed CNN/LSTM/BERT/XLM models by 1 to 3% [56]. Recent research links LLM capabilities to ASAG needs [39, 57, 58, 59, 60, 61, 62]:

- Natural Language Understanding → Handle diverse student phrasing [57]
- Semantic Similarity → Match student answers semantically to reference answers [39]
- Text Classification → Assign accurate grades [58]
- Explainability → Provide interpretable feedback [59]
- Contextual Embeddings → Capture nuanced meanings [60, 61]
- Robustness → Tolerate typos and grammatical errors [62]

2.1.5 ChatGPT in ASAG

Generative AI tools such as ChatGPT assist instructors in creating questions and supporting student comprehension [46, 63, 64]. Experimental use in grading shows efficiency gains in programming assignments [65], but reliability remains a challenge: low confidence ranges in generated answers [66], fabricated sources in literature reviews [67], and inaccurate responses in clinical domains [68]. Accuracy remains a major concern [65, 68, 69, 70, 71]. Despite these limitations, the integration of ChatGPT, LLMs, and NLP continues to grow rapidly; research publications increased from 42 in 2023 to 124 in 2024, reflecting wide adoption across disciplines.

2.1.6 BERT for Automatic Grading (ASAG)

BERT is a widely used transformer-based model for text understanding and has been applied in auto-grading systems [39, 72, 73, 23]. Across 2016–2024, the literature reports a steady rise in BERT-based ASAG studies, spanning roughly ten subject areas and numerous publication venues. The breakthrough by Devlin et al. introduced bidirectional pre-training that models both left and right context, establishing new state-of-the-art results on many NLP benchmarks [46, 24]. For ASAG, this bidirectional context lets models assess how words function both individually and collectively within a response, improving robustness to phrasing variation and enabling rubric-aligned scoring [22, 42, 52].

Why BERT works well for ASAG.

- **Contextual comprehension.** BERT conditions on full bidirectional context, enabling nuanced interpretation of student answers with respect to the question [46, 24].
- **Semantic alignment for scoring.** Fine-tuned BERT variants can compare student responses to reference answers and map representations to grades [22, 52].
- **Handling variants/synonyms.** Contextual embeddings help recognise paraphrases (e.g., “photosynthesis” vs. “the process of photosynthesising”) [22].

- **Sentence-level coherence.** BERT captures inter-sentence relations, useful when grading requires global cohesion and structure [42, 52].

2.1.7 XLNet for Automatic Grading (ASAG)

XLNet extends transformer pre-training via permutation language modelling [18], aiming to capture bidirectional dependencies without masking, and to mitigate certain pre-training biases [74]. Prior work situates XLNet as an evolution relative to BERT-style encoders, with stronger modelling of word order permutations and longer-range dependencies [23, 74]. In ASAG contexts, these properties can benefit tasks where subtle context links and multi-clause justifications matter. Empirically, domain applications such as mathematical expression re-organisation report very high accuracy (e.g., 99.80%) when using XLNet-based pipelines [75]. Although fewer ASAG papers use XLNet compared with BERT, publications from 2020–2024 span multiple disciplines and outlets, suggesting room for growth.

Why XLNet helps in ASAG.

- **Richer context capture.** Permutation-based training improves sensitivity to alternative word orders and interdependencies in student answers [23, 74].
- **Reduced pre-training bias.** The objective lessens strict left-to-right or right-to-left biases, supporting fairer treatment of diverse phrasings [74].
- **Conceptual consistency.** Stronger modelling of global context aids judgement of logical organisation and answer coherence [23].
- **Long-range reasoning.** Better handling of long-distance links can improve grading for higher-level, multi-sentence justifications [23, 75].

2.2 Computational Similarity Methods for ASAG

Computational similarity methods estimate how closely a student’s short answer matches an expected (reference) answer using NLP+ML/DL pipelines [18] that prioritise meaning over literal word overlap [76, 46, 24]. Representative systems include Arabic ASAG with the AR-ASAG dataset [77, 78, 79], rubric-aware grading tailored to question types [79], and TF-IDF features with K-Means classification [80]. In practice, similarity scores (e.g., cosine, Levenshtein) are embedded in classifiers to assign grades [24, 81, 82].

2.2.1 Method Families (with examples)

Lexical/character-based (surface similarity). Longest common substring, Jaro, Levenshtein, and n -gram indices quantify string overlap and edit distance [83, 84, 81, 85, 86, 87, 88]. These are efficient and simple but brittle under paraphrase and long texts [81].

Term/vector-based (distributional similarity). Cosine over TF-IDF or neural vectors; Web Jaccard; overlap coefficients [81, 89, 90]. Vector-based search/graphs can prune paths and speed up document-level components in ASAG pipelines [91].

Semantic (corpus/knowledge). Corpus-based measures (e.g., Normalised Google Distance, LSA) and knowledge-based measures (e.g., Resnik) target meaning beyond surface forms [92, 93, 94, 95, 96]. These can be unstable (web counts) or corpus/ontology dependent.

Sentence-level (word/structure/vector). Word-, structure-, and vector-based sentence similarities capture ordering, grammar, and dense semantics [82, 97, 93, 98, 99, 91]. Structure-based methods are accurate yet computationally heavy and language/rule specific [99, 100, 101].

2.2.2 Usage Trends and Hybrid Designs

Since 2017, DL-based similarity pipelines account for about 20% of reported systems [18], while ML remains more prevalent ($\sim 33\%$) [102, 103, 44]. Corpus-based approaches are rare ($\sim 4\%$) and usually part of hybrids [102, 103, 44]; one knowledge-based study appeared only in a corpus+ML hybrid [103]. Many works combine methods to raise accuracy or to compare alternatives [104]. Typical combinations include ML-only [105], IE+ML/DL [106], concept mapping+ML [107], and ML+DL [108]; a few use a single approach (e.g., DL-only) [109]. Three-way hybrids appear as corpus+knowledge+ML or corpus+ML+DL [103, 44]. Ensembles (e.g., stacked regression) often outperform single models [103, 45, 110]. A broader evaluation of short-answer methods provides historical context [111].

2.2.3 What Each Family Captures (and Misses)

- **Lexical/character** (LCS, Jaro, Levenshtein, n -grams): fast surface matching; sensitive to spelling and overlap; weak on paraphrase [83, 84, 81, 85, 86, 87, 88].
- **Term/vector** (cosine, Web Jaccard, overlap): distributional proximity; may be domain dependent or low-accuracy in some settings [81, 89, 90].
- **Semantic (corpus/knowledge)**: meaning-level similarity; subject to corpus/ontology coverage and stability of counts [92, 93, 94, 95, 96].
- **Sentence-level** (word/structure/vector): order, syntax, dense meaning; structure-based least used due to complexity [98, 99, 91, 100].

2.2.4 Representative Formulae and Metrics

Lexical/term examples: Levenshtein-based similarity $S(S_A, S_B) = 1 - \frac{d(S_A, S_B)}{\max(l(S_A), l(S_B))}$ [84]; cosine $S(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$ [81].

Semantic examples: NGD and Resnik information content [93, 94].

Evaluation: Accuracy, precision, recall, and F1 are commonly reported for classification-style graders [76, 112, 99, 25].

2.2.5 Known Limitations

Character-based methods can be space- or size-inefficient (recursion; small-text bias; poor scaling) [83, 86, 81, 87, 88]. Term-based methods can be domain dependent or lower in accuracy (cosine; Web Jaccard) [81, 89]. Corpus/knowledge semantic methods face visibility/stability issues and corpus-boundedness (LSA/NGD/Resnik) [92, 93, 94, 95, 96]. These challenges motivate hybrid and ensemble pipelines that mix complementary signals [102, 103, 44, 104, 45, 110].

2.3 Grading Methods Associated with ASAG Systems

Grading converts similarity or prediction outputs into interpretable outcomes for assessment [18]. Because student phrasing varies, manual marking is time-consuming and burdensome for instructors [113]. Automated grading reduces workload using multiple schemes and models [81, 112, 111, 23, 107].

2.3.1 Taxonomy of Grading Schemes

Method	Description
2-way	Correct / Incorrect [111]
3-way	Correct / Partially Correct / Incorrect [111]
5-way	Correct / Partially Correct / Contradictory / Irrelevant / Non-domain [111]
Points (0–5)	Ordinal score band, e.g., 0 (lowest) to 5 (highest) [76, 114, 115, 116, 117]
Hybrid	Combination of two or more of the above [107, 118]

2.3.2 Observed Usage and Performance

Empirical results [18] show that binary (2-way) setups can be highly effective: Wu and Yeh report a correct/in-correct system with 91% accuracy [112], and Marvaniya et al. find 2-way classification yields the best macro-averaged performance versus 3- and 5-way alternatives [107]. Prior reviews catalogue 2-/3-/5-way variants and their trade-offs [111]. Points-based scoring (e.g., 0–5) is the most frequently used scheme, whereas pure 5-way is rare; several studies blend 5-way into hybrid designs to boost performance [76, 114, 115, 116, 117, 107, 118]. Overall, six studies explicitly use hybrid grading to evaluate efficiency and robustness [107, 118].

From similarity to grades. Most systems first compute a similarity or model score [18] and then map it to a grade band or point value. Reported usage shares indicate points/scoring dominate (about 46%), with only a single project using exclusively 5-way, five projects merging 5-way with others, and multiple studies adopting hybrid mappings to suit task/rubric needs [107, 118, 76, 114, 115, 116, 117].

2.3.3 Pedagogical Role

Grading outputs support both summative and formative assessment. With point-based designs, scores can track progress, guide feedback, and motivate self-regulated learning, especially in quizzes and online training [28, 29, 119, 120, 121].

2.3.4 Representative Study Choices

Point-based grading is common in practice [122, 123, 124]. Some papers compare multiple schemes side-by-side (2-/3-/5-way) [107, 118]; others implement 5-way specifically [45]. Several works use only 3-way [19, 42, 79, 106]. Selection is often driven by rubric granularity, domain conventions, and dataset label design.

2.4 Datasets Used in ASAG Studies

ASAG research draws on both *pilot/operational* datasets collected from real settings and *public* benchmarks. Many studies start [18] with pilot data to demonstrate real-world viability [114, 115, 125, 126, 127, 128]. Widely used public sets include the Mohler/Extended Texas Computer Science datasets [129], Kaggle collections [130] (frequently reused [20, 45, 110]), and SciEntBank [112, 103, 23]. Recent large-scale industry data have also appeared (e.g., Sung et al.) covering *Physiology of Behavior (Phy)*, *American Government (Gov)*, *Psychology—Human Development (Psy-I)*, and *Abnormal Psychology (Psy-II)* [131]. Multilingual resources are emerging, such as the Arabic AR-ASAG dataset [77].

2.4.1 Usage Patterns

Across the reviewed literature, pilot datasets are the most common. A representative breakdown is shown below [18]:

- **Pilot datasets:** ~ 54% (28 approaches) [114, 115, 125, 126, 127, 128].
- **SemEval datasets:** 7 approaches [132].
- **Mohler/Extended Texas:** 6 approaches [129].
- **Cairo University:** 2 approaches [78].
- **Kaggle:** 2 approaches [130].
- **Project-specific (individual):** 5 approaches [19, 112, 102, 103].
- **Large-scale industrial:** 1 approach [36].

In addition to English-only corpora, non-English datasets have been used (e.g., AR-ASAG in Arabic) [77].

2.4.2 Publicly Available Datasets (examples)

Several datasets are commonly cited as accessible baselines for ASAG experiments:

- **Texas Computer Science and Extended Texas (Mohler)** [133, 129].
- **Cairo University** [134].
- **SemEval 2013** short-answer resources [132].
- **Kaggle** task collections reused across studies [130, 20, 45, 110].
- **SciEntBank** science QA corpus [112, 103, 23].

2.4.3 Why Dataset Quality Matters

Reliable ASAG evaluation depends on high-quality, rubric-aligned datasets—not only on algorithms (NLP/M-L/DL). Robust measurement of efficiency and accuracy requires coverage of domains, lengths (short phrases to paragraph/essay), and languages, together with defensible labels for formative and summative use [114, 115, 125, 126, 127, 128, 28, 29, 119, 120, 121].

Category	Examples / References	Reported Usage
Pilot / operational	Institution- or course-specific pilots [114, 115, 125, 126, 127, 128]	~54% (28)
SemEval	SemEval 2013 short-answer resources [132]	7 approaches
Mohler / Extended Texas	Texas CS / Extended Texas [129, 133]	6 approaches
Cairo University	Arabic/education datasets [78, 134]	2 approaches
Kaggle	Public task collections [130]	2 approaches
Project-specific	In-house research datasets [19, 112, 102, 103]	5 approaches
Industrial (large-scale)	Multi-domain production data [131, 36]	1 approach
Multilingual	AR-ASAG (Arabic) [77]	–

2.5 Research Gaps

Despite decades of progress in automated assessment, existing approaches to **Automatic Grading** remain limited in their ability to address the complexity and diversity of real-world educational assessments. Most current systems are designed for **narrow textual responses**, typically short answers or brief essays averaging less than a few hundred words.

A fundamental gap in the current literature is the **inadequate integration of rubrics** into the automated grading process. Existing methods often treat rubrics as unstructured lists of criteria rather than as hierarchical, rule-based frameworks that encode the intended logic of assessment. This simplification neglects dependencies among criteria, variable scoring rules, and the hierarchical relationships that educators explicitly design into rubrics to ensure fairness and transparency. As a result, these systems struggle to produce **consistent, rubric-aligned decisions** and fail to offer meaningful explanations that link scores to the structure and intent of the rubric. The lack of **rule- and rubric-based explainability with structural reasoning** further limits the pedagogical usefulness of automated systems, as they cannot provide teachers or students with interpretable rationales or targeted feedback for improvement.

Another critical limitation lies in the **restricted generalizability** of existing models across subjects and assessment types. Most ASAG approaches are **domain-dependent**, requiring substantial retraining or fine-tuning whenever applied to a new exam or discipline. This dependence on task-specific labeled data not only increases development cost and time but also hinders scalability and equitable deployment across diverse educational contexts. Moreover, models trained in one domain often exhibit degraded performance when applied to others, revealing a lack of robustness in capturing domain-independent linguistic and conceptual patterns. Together, these limitations highlight the urgent need for a **scalable, subject-agnostic, and rubric-aligned framework** that supports a wide spectrum of textual responses, from short answers to multi-page essays—while maintaining explainability, fairness, and operational reliability in real-world educational environments.

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Algorithm 4 Answer Scoring Based on RKT and ACT

Require: RKT, ACT, Matching-Limit

Ensure: Score and Reason for each node

```
1: Leaf Node Scoring:
2: for each leaf  $N$  in RKT do
3:   for each leaf  $M$  in ACT do
4:      $MP \leftarrow LLM\_SSM(N.criteria, M.goal, M.abstarctive\_summary)$ 
5:     if  $MP > \text{Matching-Limit}$  then
6:        $[S, R] \leftarrow LLM\_SSR(M.Text, N.Criteria\_Simplified\_version, N.la)$ 
7:        $N.Score[N.ID, M.ID] \leftarrow S, \quad N.Reason[N.ID, M.ID] \leftarrow R$ 
8:       if  $N.criteria\_type \in \{WHOLE, STRUCTRAL\}$  then
9:         for each leaf  $T$  in  $N.influence\_section$  do
10:           $[S, R] \leftarrow LLM\_SSR(\text{all } T.Text, N.Criteria\_Simplified\_version, N.la)$ 
11:           $N.Score[N.ID, T.ID] \leftarrow S, \quad N.Reason[N.ID, T.ID] \leftarrow R$ 
12:        end for
13:      end if
14:    end if
15:  end for
16:   $T^* \leftarrow \arg \max_T N.Score[N.ID, T.ID]$ 
17:  for each ancestor  $P$  of  $T^*$  in ACT do
18:     $N.Score[N.ID, P.ID] \leftarrow N.Score[N.ID, T^*.ID]$ 
19:  end for
20: end for
21: Internal Node Aggregation:
22: for each internal node  $N$  in RKT do
23:   for each child  $C$  of  $N$  do
24:     for each  $L$  in RKT, each leaf  $T$  in ACT do
25:        $N.\{Score, Reason, Matching\_Percentage\}[L.ID, T.ID] \leftarrow$ 
26:          $C.\{Score, Reason, Matching\_Percentage\}[L.ID, T.ID]$ 
27:     end for
28:   end for
29:   for each leaf  $T$  in ACT do
30:      $N.Score[N.ID, T.ID] \leftarrow SAF(\{C.Score[C.ID, T.ID]\}, N.relationship\_type, N.criteria\_type)$ 
31:      $N.Reason[N.ID, T.ID] \leftarrow RAF(\{C.Reason[C.ID, T.ID]\}, N.relationship\_type, N.criteria\_type)$ 
32:      $N.Matching\_Percentage[N.ID, T.ID] \leftarrow \max_C N.Matching\_Percentage[C.ID, T.ID]$ 
33:   end for
34:   for each internal ACT node  $F$  do
35:      $Q \leftarrow \arg \max_Q \{N.Score[N.ID, Q.ID] \mid Q \text{ is child of } F\}$ 
36:      $N.Score[N.ID, F.ID] \leftarrow N.Score[N.ID, Q.ID]$ 
37:   end for
38: end for
39:  $=0$ 
```
